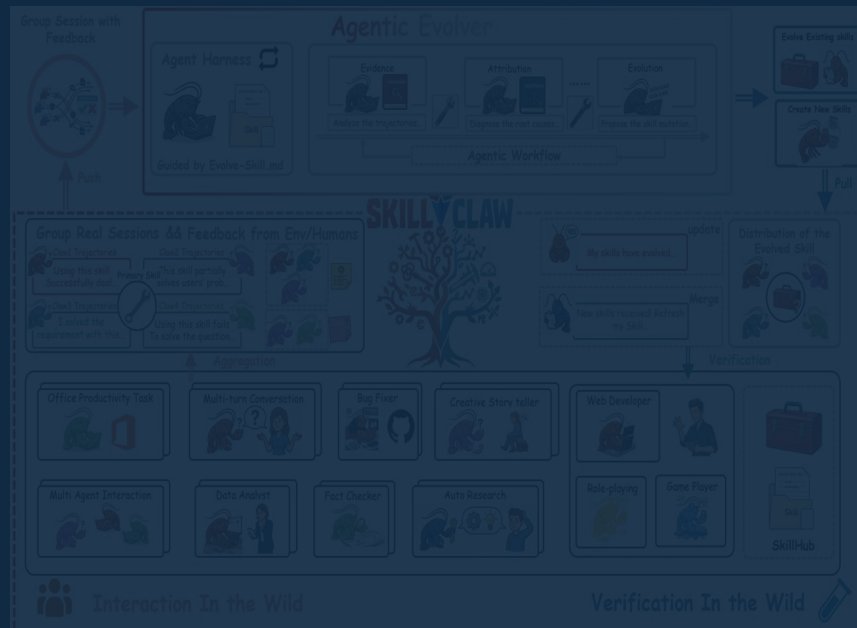


SkillClaw

Collective Skill Evolution for Multi-User LLM Agent Ecosystems

Runtime interaction trajectories become shared evidence for continuously evolving agent skills.

Source: arxiv.org/abs/2604.08377



Multi-user interaction → Session collection → Skill evolution → Skill synchronization

Agent Skills Remain Static After Deployment

Users repeatedly rediscover failures and recoveries instead of accumulating system-level knowledge.

Static skill paradigm

- Manual install from centralized hub
- Runtime solutions rarely persist
- Similar failures recur across users
- Memory methods stay instance-tied



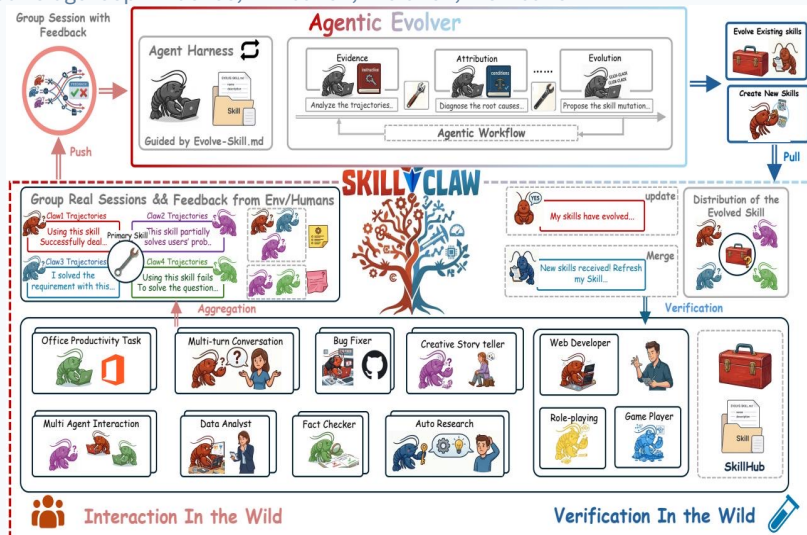
Evolving skill paradigm

- Trajectories preserve causal chains
- Cross-user evidence is grouped by skill
- Verified mutations update shared repository
- All agents synchronize improved skills

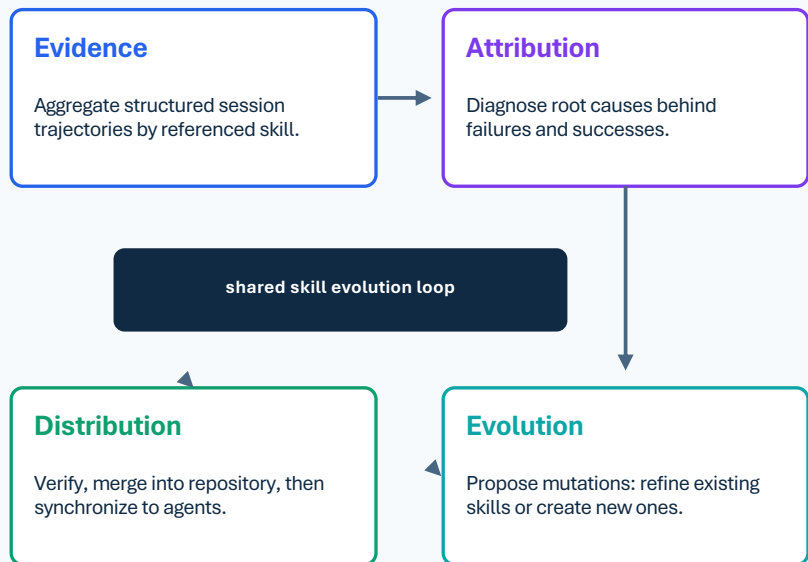
SkillClaw closes this gap by turning usage into reusable skill updates.

Closed-Loop Pipeline: Multi-User Interaction → Shared Skill Evolution

Four-stage loop: Evidence, Attribution, Evolution, Distribution.

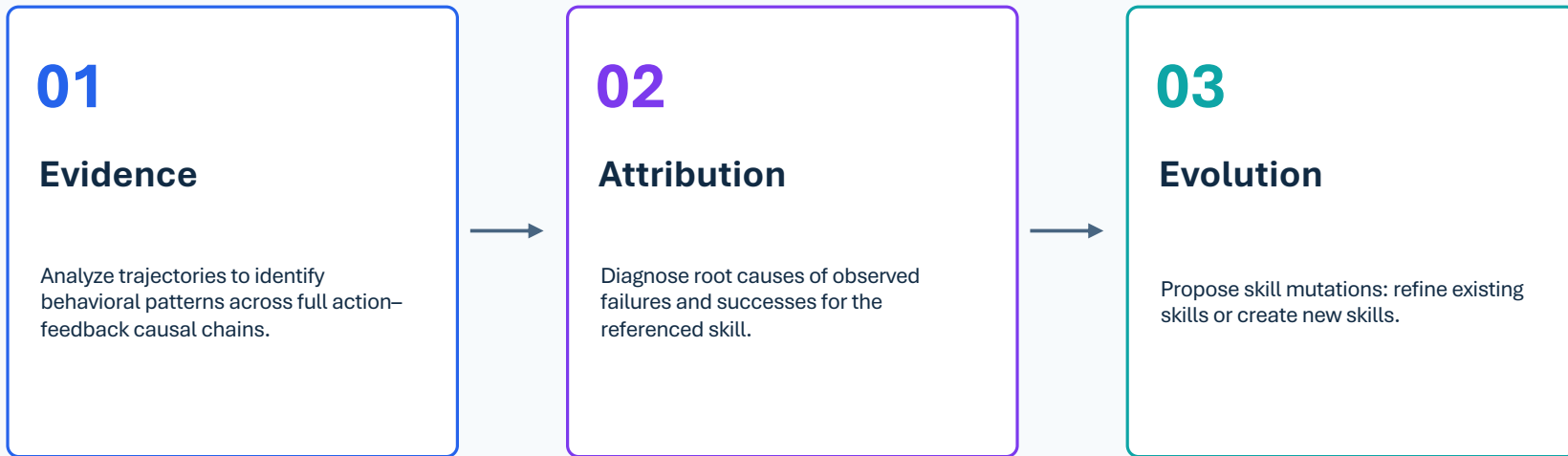


Architecture overview figure anchors the centralized evolution loop: users generate sessions; SkillClaw aggregates, evolves, verifies, and synchronizes skills.



Agentic Evolver: Evidence → Attribution → Evolution

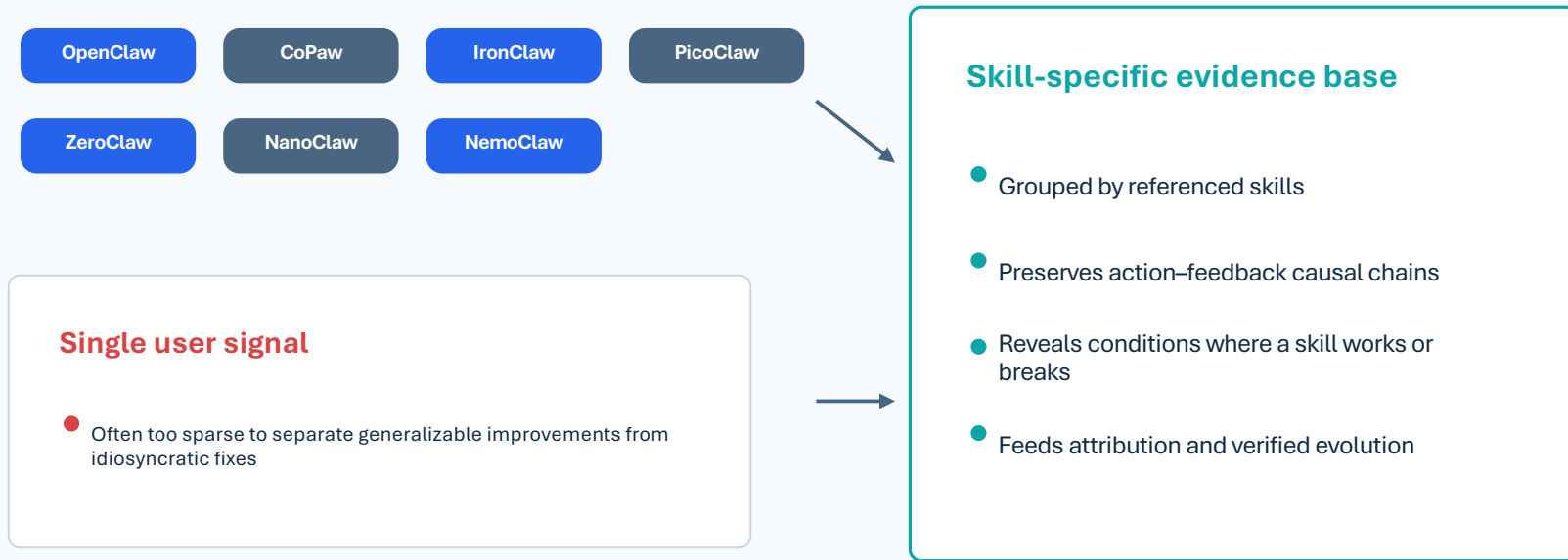
Open-ended agent reasoning over interaction evidence, not predefined update rules.



Why agentic? The evolver can reason over heterogeneous evidence and propose context-aware mutations without a fixed rule template.

Cross-User Trajectory Aggregation Exposes Behavioral Boundaries

Compatible Claw-style systems contribute complementary success and failure signals for the same skill.



Aggregation turns repeated use into a boundary map: success patterns + failure modes.

Case Study: Multimodal Creative Pipeline — 6 Days of Attempts

Only 1 of 6 candidates was accepted, illustrating conservative verification that prevents regressions.

Date	Candidate / Change	Outcome	Rationale / next action
Day 1	validate-tmp-workspace-inputs	Accepted	Checks /tmp_workspace inputs and environment; upgrade to new best pool.
Day 2	multimodal-input-validation-and-setup	Rejected	Multimodal input validation and output init; keep current best pool.
Day 3	multimodal-creative-task-pipeline	Rejected	Pipeline extracts from PDF/video/image; keep current best pool.
Day 4	creative pipeline (improved)	Rejected	Added image classification, visual generation, structured validation; keep pool.
Day 5	pipeline + validate-required-input-files	Rejected	Creative pipeline plus per-file fail-fast validation; keep pool.
Day 6	creative pipeline candidate	Rejected	Extended PDF-to-poster / document-to-visual paths; not admitted.

1 / 6 accepted

Case Study: Git Push Auth-Fallback — Iterative Refinement

Here 3 of 6 days yielded accepted updates before the skill plateaued.

Date	Candidate / Change	Outcome	Rationale / next action
Day 1	git-push-with-auth-fallback	Accepted	Safe fallback instead of blocking on push failure; add to Safety best pool.
Day 2	git-push-with-auth-fallback	Accepted	Unified patch/bundle filenames and reduced inconsistency; keep updated pool.
Day 3	fallback + git-clone-to-directory	Accepted	Auth-alternative paths, secrets audit, subshell fixes; keep current pool.
Day 4	same-pool retest	No update	Validator confirmed current pool as best; continue same best pool.
Day 5	git-push-with-auth-fallback	Rejected	Added “push hang treated as auth failure”; no improvement.
Day 6	git-push-with-auth-fallback	Rejected	Added identity config and filename consistency; not admitted.

3 / 6 accepted updates

Key Takeaways: Three Properties That Distinguish SkillClaw

Implications for continuously improving agent systems.

01

Collective evolution

Individual interactions improve a shared, continuously improving skill ecosystem.

02

Fully automatic

Runtime interaction drives skill evolution without manual curation or explicit user intervention.

03

Agentic evolution paradigm

Open-ended reasoning over evidence produces updates instead of predefined rules.

Design implication: treat deployed skills as living system components that improve through verified collective experience.